

NONRESPONSE BIAS ON DIMENSIONS OF POLITICAL ACTIVITY AMONGST POLITICAL ELITES

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Virtually all political surveys of citizens' political behavior suffer from unit nonresponse. Typically, researchers' analyses implicitly assume that the nonrespondents do not differ systematically from the respondents on dimensions of political activity (King, Honaker, Joseph, & Scheve, 2001). For example, regressions fit with only respondents assume that the model for respondents applies equally to nonrespondents. When such similarity assumptions are not valid, conclusions from standard statistical analyses can lead to inaccurate inferences. The potential for nonresponse bias is especially worrisome in surveys of political elites, who sometimes have low response rates (e.g., McClosky, Hoffmann, & O'Hara, 1960; Hedges, 1984; Whitehead, Blankenship, & Wright, 1999; Goldstein, 2003).

Low response rates do not indicate bias in estimates *per se* (Groves, 2006). To assess the potential for bias, the survey researcher needs external information gathered independently of response decisions. For example, in some surveys it is possible to check the similarity assumptions partially by comparing known demographic data for the nonrespondents and respondents. Brehm (1993) compares the results of the General Social Survey (GSS) to the Current Population Studies (CPS) and finds that proportionately fewer youth, whites, and males respond to the GSS. Abraham, Maitland, and Bianchi (2006) perform a similar analysis in analyzing the American Time Use Survey.

While such comparisons are possible for demographic variables, they are made less frequently for substantive variables asked in surveys, like political views or financial matters. The problem is that there are no population-wide data on political views or behaviors to provide a baseline for comparison. Additionally, such views (and the topics of interest) can change rapidly in a short period of time. That said, some efforts have been made to compare respondents' and nonrespondents' political views and actions. Generally, these involve face-to-face or phone interviews that entail asking would-be nonrespondents to answer just a few questions (Brehm 1993; Voogt & Van Kempen, 2002; Voogt & Saris, 2003). Such projects suggest that nonrespondents to political surveys are generally less politically active, show less political interest, and score lower on measures of political efficacy than do respondents. Since many political questions addressed to elites concern political interest and involvement, these findings raise concerns that low response rates in surveys of elites may introduce nonresponse bias.

Unfortunately, there are few comparable studies of nonrespondents conducted amongst more elite populations even though we often know much more about elite nonrespondents. The one outstanding exception of which we are aware relates to the Survey of Consumer Finances (SCF) funded by the United States Federal Reserve. Several authors (cf. Montalto & Sung, 1996; Kennickell, 1999; Kennickell & Woodburn, 1999) have used external data sources, such as tax records, to look at traits of nonrespondents in the top wealth stratum. In this instance, researchers were able to examine nonrespondents along dimensions of economic traits such as wealth and income. However, to our knowledge such a procedure has never been followed for surveys of political elites along dimensions of political activity. Even in surveys with relatively low rates of nonresponse, little is understood about whether and how respondents differ from nonrespondents along dimensions of political activity within more elite samples (Roback, 1975; Sandys & McGarrell, 1994; Brown, Powell, & Wilcox, 1995; Golden, 2000; Herrnson, 2000; Francia, Herrnson, Green, Powell, & Wilcox, 2003; Miller, 2007). In fact, in some studies of elites, traditional response rates are not reported at all (e.g., Wittkopf & Maggiotto, 1983; Dalton, 1985; Dodson, 1990).

In this article, we utilize a data source that makes it possible to compare respondents and nonrespondents on several dimensions of political activity—some of which have not, to our knowledge, previously been examined. The data come from a survey of campaign donors to presidential candidates in the 2000 US national elections. The US Federal Election Commission (FEC) keeps and publicizes reports about campaign donors, making it possible to compare respondents and nonrespondents.

DATA

Following the 2000 United States presidential election, a team of researchers led by Michael Munger, and including Alexandra Cooper, John Green, Mark Rozell, and Clyde Wilcox, conducted a mail survey of donors who had contributed to any of eight presidential candidates. These included two Democrats (Bill Bradley and Al Gore), five Republicans (Gary Bauer, George Bush, Bob Dole, Steve Forbes, and John McCain), and Green candidate Ralph Nader. The sample was stratified by candidate; each candidate's sample included approximately 300 donors sampled from public lists of donors maintained by the Federal Election Commission. Strata for Dole, Nader, and Bush were over-sampled for substantive reasons.

The 10-page survey instrument contained items designed to collect data on donor demographics; on political attributes, attitudes, and activities; on how the donors were solicited; and on how much, how often, and to whom they gave. It was sent directly to sampled donors via first class mail. Each envelope was addressed to a specific donor and included a personalized cover letter explaining the project. All letters were individually signed by hand. Business-reply return envelopes were provided with these materials. Approximately one month after the initial mailing, a follow-up request was sent to all sample members who had not returned a response. The text of the cover letter was changed to reflect a repeat request; otherwise, the contents were identical to the first mailing. One additional mailing was done for all remaining

nonrespondents, except those in Nader's stratum.¹ Participants were given the option of responding anonymously to the survey by erasing the unique identifying number written in pencil on their surveys. Eight percent of the total sample chose to respond anonymously, with the highest rates of anonymous respondents from the Bauer (11 percent) and Forbes (10 percent) strata, and the lowest rates of anonymous response from the Gore (5 percent), McCain (6 percent), and Bush (6 percent) strata. The three waves were sent over the period from August to November of 2001.

To understand our comparison of respondents and nonrespondents within this sample, it is useful to have some basic information about the elite nature of campaign donors, who are quite different from 'average' Americans. The data collected from respondents to this survey reflect this fact. The median family income of the respondents is between \$100,000 and \$240,000, and 15 percent of respondents made more than \$500,000. Over 80 percent of respondents have college degrees, and 43 percent have post-graduate degrees. The respondents are deeply involved in the political process: 7.2 percent report having testified before Congress and over 15 percent report having spoken with the president.

Our comparison of respondents and nonrespondents is possible because campaigns must publicly disclose information about individual donors who give more than \$200 during the election cycle. For this group of donors, it is thus possible to compare respondents and nonrespondents along dimensions that are important to researchers in political science and public opinion. These include the sizes, numbers, and dates of contributions, the donors' targeted presidential candidates and employment status, and the numbers of contributions made to other federal campaigns, party organizations, and political action committees (PACs). Not only were we able to locate data on presidential donation, but also could find data on any contribution above \$200 to any political candidate or group required to file public reports to the FEC. While these data are certainly not perfect, campaigns often spend a great deal of time and effort maintaining this database, as it serves as the basis for fund raising in future cycles. For the surveys included in this study, only 5.89 percent of the surveys we mailed were returned as undeliverable with no forwarding address.

RESULTS

We use AAPOR/WAPOR Response Rate 2 (AAPOR, 2000) to define nonresponse. Out of the 2,594 sampled donors, 645 responded to the initial mailing, 156 to the second mailing, and 67 to the third mailing. The characteristics of respondents did not differ substantially by wave. Therefore, we combine all respondents in one category. Individuals who chose to respond anonymously are necessarily excluded from this analysis, as we have no way to link their data with the data included in the FEC database upon which we base our inferences.

¹ The response rate from Nader donors exceeded 40 percent after the second mailing; limited financial resources led the research team to decide not to recontact this stratum. However, only 67 individuals responded in the third wave across all subsamples. We detected no substantial differences across waves, and therefore feel that this does not unduly reduce the generality of our inferences.

TABLE 1 Sample sizes and response rates for full sample and candidate subsamples

	<i>Full sample</i>	<i>Bauer</i>	<i>Forbes</i>	<i>Gore</i>	<i>McCain</i>	<i>Nader</i>	<i>Bradley</i>	<i>Dole</i>	<i>Bush</i>
Sample size	2594	296	281	295	291	396	296	390	349
Responded (%)	33.5	32.4	33.1	34.2	37.5	45.5	31.8	26.2	26.7

Table 1 displays information about the effective sample size and response rates for the (unweighted) sample as well as for each subpopulation. Response rates are around 33 percent for donors to Bauer, Forbes, Gore, McCain, and Bradley. Donors to Nader responded with the highest frequency (45 percent), despite the fact that they did not receive a third mailing, and donors to Bush and Dole responded with lowest frequencies (26 percent).

Table 2 displays summary statistics for nonrespondents and respondents for several variables that help characterize their level of political activity. These include (a) the dollar amount of the sum of all the contributions they made during the 2000 election cycle, (b) the average dollar amount per contribution, (c) the number of unique recipients to whom they donated, (d) the total number of contributions they made during the cycle, and (e) the date of their first contribution measured in days since January 1, 1999. Note that all statistics for the full sample are weighted to reflect the different probabilities of being sampled from each subsample.

Table 3 includes several dichotomous variables that help to further characterize respondents and nonrespondents. These characteristics include (a) whether or not the individual reported being employed in the workforce, (b) whether or not the individual gave to a congressional candidate, (c) a party organization, or (d) a PAC, and (e) whether or not the individual gave the maximum amount (\$200 – \$1,000 to the campaign itself, and an additional \$1,000 to the candidate's compliance fund) allowed under federal elections laws to any specific presidential candidate. We therefore were able to compare respondents and nonrespondents along dimensions of political participation in considerable detail. To our knowledge, such data on nonrespondents have not been collected in previous political surveys.

EMPLOYMENT

Overall, 72.5 percent of respondents and 77.7 percent of nonrespondents were employed in the market, a difference that is statistically significant ($p = .06$). For all candidates except Dole and Bauer, the employed comprise a smaller percentage of respondents than nonrespondents.²

²The FEC data did not include employment information for 290 individuals. The FEC requires that campaigns include information about the employer of their donors in quarterly reports. Although it is not always easy to disaggregate individuals by their exact jobs, it is almost always straightforward to identify individuals who are employed by a company and those who are retired, home-makers, or otherwise not employed in the market.

TABLE 2 Contribution characteristics for full sample and candidate subsamples

	<i>Full sample^a</i>	<i>Bauer</i>	<i>Forbes</i>	<i>Gore</i>	<i>McCain</i>	<i>Nader</i>	<i>Bradley</i>	<i>Dole</i>	<i>Bush</i>
<i>Sum of all contributions</i>									
Respondents	\$3358* (362)	\$1315* (257)	\$10936 (5956)	\$4870 (1295)	\$1511* (250)	\$804* (152)	\$2902* (497)	\$4749 (718)	\$3225 (476)
Nonrespondents	\$5548 (698)	\$961 (125)	\$6523 (1292)	\$6869 (1935)	\$3768 (1142)	\$4427 (1607)	\$10305 (3215)	\$11171 (4442)	\$3939 (673)
<i>Average contribution</i>									
Respondents	\$622* (34)	\$485* (33)	\$693* (146)	\$619* (33)	\$508* (29)	\$416* (21)	\$624* (49)	\$685* (43)	\$674* (79)
Nonrespondents	\$888 (37)	\$418 (23)	\$873 (82)	\$885 (81)	\$732 (76)	\$648 (70)	\$1052 (112)	\$932 (74)	\$900 (58)
<i>Total number of recipients</i>									
Respondents	3.0 (0.16)	1.7* (0.15)	4.5 (0.61)	3.4 (0.39)	2.1 (0.27)	1.4* (0.11)	3.0 (0.35)	3.9 (0.42)	3.1* (0.29)
Nonrespondents	2.8 (0.11)	1.4 (0.08)	3.5 (0.32)	3.2 (0.21)	2.3 (0.22)	2.3 (0.25)	3.2 (0.28)	4.1 (0.36)	2.6 (0.18)
<i>Total number of contributions</i>									
Respondents	4.5 (0.31)	2.3 (0.23)	8.1* (1.20)	5.5 (0.79)	3.0 (0.44)	1.6* (0.14)	3.9 (0.51)	5.9 (0.70)	4.8* (0.60)
Nonrespondents	4.1 (0.19)	2.1 (0.15)	5.8 (0.81)	4.8 (0.37)	3.3 (0.39)	3.1 (0.41)	4.6 (0.52)	6.8 (0.69)	3.8 (0.31)
<i>Date of first contribution^b</i>									
Respondents	233* (8.5)	153 (11.2)	156 (9.8)	235 (18.8)	299 (13.1)	544* (8.0)	192 (12.4)	136 (7.5)	226 (17.1)
Nonrespondents	209 (5.2)	165 (7.9)	164 (6.0)	202 (11.9)	286 (10.5)	491 (11.9)	176 (7.4)	144 (4.2)	208 (9.0)

Standard errors in parentheses.

^aFull sample statistics weighted for probability of being sampled.^bDays since January 1, 1999.* $p \leq .10$ for two-sided hypothesis test for differences in means between respondents and nonrespondents.

CONTRIBUTION SIZE

Nonrespondents differ most conspicuously from respondents in the sizes of their contributions. Combining all candidates' data, the weighted sum of all donations during the cycle and the weighted per donation average for respondents are respectively about \$2,190 lower and \$266 lower than the corresponding averages for nonrespondents. This ordering holds for all candidates except Bauer and Forbes. Two-sided significance tests performed after taking logarithms of the sums and averages result in p -values of .05 or less in most cases. There were no statistically significant differences in the logged total contributions for the Forbes, Gore, Dole, and Bush subsamples. The log transformation was considered necessary due to highly right-skewed distributions. Thus, in contrast to

TABLE 3 Donor characteristics for full sample and candidate subsamples (percentages)

	<i>Full sample^a</i>	<i>Bauer</i>	<i>Forbes</i>	<i>Gore</i>	<i>McCain</i>	<i>Nader</i>	<i>Bradley</i>	<i>Dole</i>	<i>Bush</i>
<i>Currently employed</i>									
Respondents	72.5*	59.4*	55.6	74.2	64.7	62.5*	79.6	81.4*	73.3
Nonrespondents	77.7	46.5	64.0	76.6	72.5	72.4	83.2	70.9	79.9
<i>Congressional donor</i>									
Respondents	39.2	18.8*	46.2	45.5	24.8	8.9*	39.4	46.1	43.0
Nonrespondents	36.7	11.5	43.1	51.6	30.2	21.3	32.2	45.1	35.2
<i>Party donor</i>									
Respondents	29.9	10.4*	43.0*	35.6	9.2	4.4*	18.1*	29.4	40.9*
Nonrespondents	26.4	5.0	27.7	31.4	13.2	10.2	29.7	34.7	27.3
<i>PAC donor</i>									
Respondents	18.1	15.6*	19.4	19.8	7.3	3.3*	18.1	22.6	21.5
Nonrespondents	15.6	9.0	14.4	16.0	8.2	11.6	17.8	16.7	16.8
<i>Maxed out to a presidential candidate</i>									
Respondents	8.1*	2.1	6.5	14.9	0.9*	2.2*	3.2*	16.7	9.7
Nonrespondents	12.7	1.0	7.5	18.6	5.5	9.7	13.4	13.9	12.9

^aFull sample statistics weighted for probability of being sampled.

* $p \leq .10$ for two-sided hypothesis test for differences in percentages between respondents and nonrespondents.

previous findings in surveys of the mass electorate, we found that respondents scored lower on measures of political participation than non-respondents. Finally, respondents also are generally less likely to give the maximum allowable amount to a presidential campaign, although this trend varies by candidate.³

DONATION PATTERNS

On average, nonrespondents make roughly the same number of donations and give to the same number of unique recipients as do respondents. Response is not strongly related to propensity to donate to congressional candidates, parties, or PACs. Respondents do donate later on average, than do nonrespondents. The 95 percent confidence interval for the average time of first donation (measured in days since January 1, 1999) is [216, 249] for respondents, contrasted with [199, 219] for nonrespondents.

³ A helpful reviewer noted that the lower levels of response amongst individuals who gave large sums may be related to a curvilinear relationship between income and responses. This would imply lower response rates at the upper and lower ends of the income distribution. This appears to be a very plausible alternative theory, but it is difficult to test without direct data on income levels for both respondents and non-respondents. To the extent that donation amount is correlated with income, however, this should be considered strongly as an alternative hypothesis.

CANDIDATE STRATUM

Respondents and nonrespondents within each stratum are fairly similar except on the variables discussed above. The exception to this overall pattern is the Nader stratum, within which there are statistically significant, and in some instances dramatic, differences between respondents and nonrespondents on almost every dimension. Nader respondents gave substantially less money to fewer candidates and at fewer levels (congressional, party, PACs) than Nader nonrespondents.

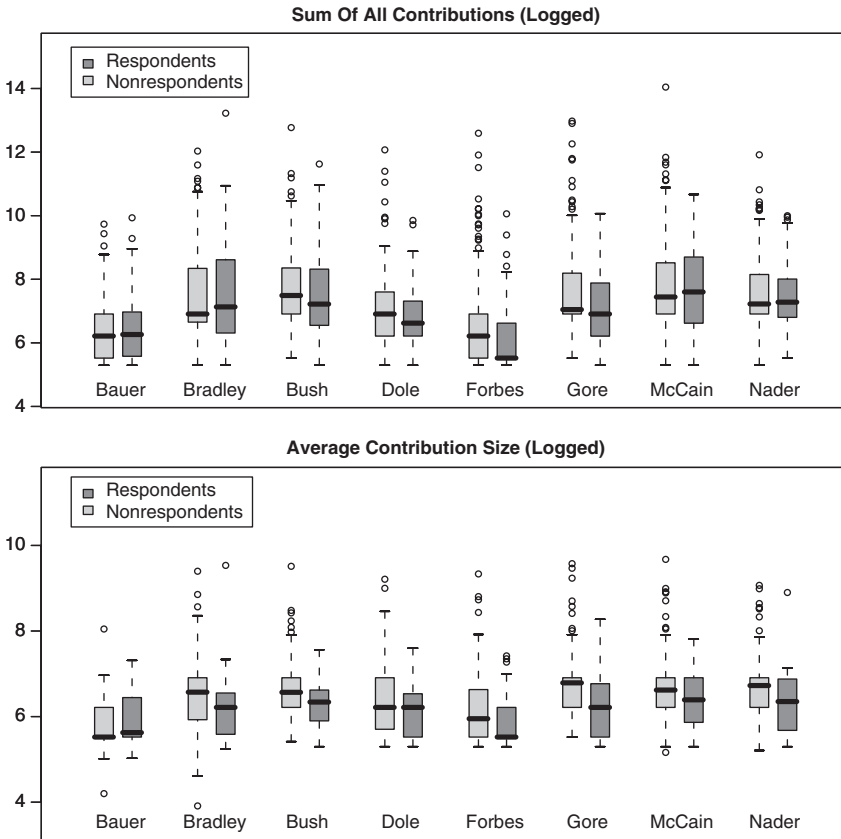
DISCUSSION

Significant problems for the analyses of political data lie in the potentially different levels of political activity between respondents and nonrespondents. Indeed, measures of political interest and activity have been a particular focus of researchers of nonresponse bias (Voogt & Saris, 2003). Respondents gave less money on average per contribution and overall during the 2000 cycle than nonrespondents, suggesting that the respondents were *less* involved in contributing money to candidates and organizations that participate in federal campaigns (i.e., PACs) than nonrespondents.

In addition, nonrespondents tended to give their first contribution prior to respondents. Political science recognizes that early donors to the political process often have substantially disproportionate influence over outcomes and that these donors are often more integrated into the system than later donors (Biersack, Herrnson, & Wilcox, 1993). The long time during which presidential candidates, in particular, raise funds leads many observers of presidential election politics to judge 'early money' as more valuable than 'late money.' Thus, the finding that nonrespondents gave somewhat earlier may imply that the nonrespondents were more integrated in the presidential campaign financing process than respondents. Finally, although the respondents' and nonrespondents' numbers and targets of donations do not differ appreciably, nonrespondents were more likely to give the maximum donation amount to a presidential candidate.

Each of these findings suggest that respondents tended to participate *less* in politics as measured by donation size and tended to be less fully engaged in the political process. Although this would be a surprising result for a survey of the mass electorate, a plausible explanation is available by thinking of these respondents as political elites. As shown in the box plots of the size variables in Figure 1, the respondents and nonrespondents differ in the upper tail of the size distribution: comparatively few of the very highest amount donors respond to the survey. Individuals in the tails of the distribution are significantly less likely to respond to the survey. Of the 36 individuals who gave more than \$50,000 to political causes and candidates in the 2000 cycle only four responded. Thus, although 33.5 percent of the (unweighted) overall population responded, only 11 percent of these outliers did so. The importance of the tails of the distribution in driving these differences can be demonstrated by noting that the median *total contribution* for *both* respondents and nonrespondents is \$1,000.

FIGURE 1 Box-plots of total contributions and average contributions by subsample



NOTE: Bold line in the center of each box-plot is the median score within that sub-population. Boxes contain from the 25th to 75th percentile. The whiskers extend to 1.5 times the inter-quartile range. Points represent outliers.

Meanwhile, the median average donation for respondents is \$500, while the median rises only to \$650 for nonrespondents—a relatively modest (but statistically significant) difference of little theoretical import. So, while respondents and nonrespondents differ in a statistical sense, these differences appear to be largely driven by outliers rather than by overall trends, and the substantive differences are more modest.

We suspect the asymmetries between respondents and nonrespondents in this survey would be viewed as not too worrisome by most researchers. The differences in the average summed contributions are small compared to the standard deviations in the overall sample and result primarily from influential outliers. The average date of contribution for both groups fall well before the Iowa caucuses or New Hampshire primary. However, some differences cannot be dismissed. Nader supporters are overrepresented, and Bush and Dole supporters, underrepresented, among the respondents. Individuals in the tails of the distribution are significantly less likely to respond to the survey.

CONCLUSION

Thus far, we have reviewed empirical differences between nonrespondents and respondents. We find that there do appear to be differences along dimensions of political activity. In particular, individuals who participate at extremely high rates in terms of frequency and total contributions responded at a lower rate. The remaining differences between respondents and nonrespondents, although statistically significant, do not appear to differ greatly in a substantive sense. Additionally, to the extent that they differ we again find that nonrespondents participated at somewhat higher levels as measured by contribution amounts.

What can account for differences between respondents and nonrespondents in this elite sample? Put in another way, are there any individual traits that might correlate with both the decision to respond and the differential traits we observe among respondents and non-respondents? Extrapolating from these data to offer an answer to these questions necessarily involves some conjecture due to obvious data constraints. After all, we have no measures of political attitudes, opinions, or even many demographic traits such as education and income for non-respondents. That said, we feel that a plausible theory can be proposed here based on the above findings that may inform future studies.

Many researchers of political participation posit that donors contribute for some combination of two reasons. One is that they hope to influence the outcome of an election—to provide needed assistance to a candidate with whom they agree. Another is that they may hope to gain the favor of an office-seeker by supporting him or her—and ideally, to gain access to that candidate once he or she is in office (cf. Verba & Nie, 1972; Wilson, 1973; Welch, 1974; Verba, Lehman Schlozman, & Brady, 1995).

Of course, these motives are not mutually exclusive, yet, *ceteris paribus*, we might expect access-seekers to give larger amounts than outcome-oriented donors, in hopes of drawing the candidate's attention to themselves (Francia et al., 2003). Additionally, access-seekers may be less ideologically driven—focused on identifying likely victors rather than on forwarding a broader political agenda (Brown et al., 1995; Francia et al., 2003). Thus, it may seem plausible to assert that respondents tend to fall more into the ideological-driven category of participants, while nonrespondents may be more motivated by access-seeking.

This is a subtly different conclusion than previous researchers have reached about the potential dangers of unit nonresponse bias along dimensions of political participation (Brehm, 1993; Groves & Couper, 1998). Previous findings have largely focused on behavior differences such as social involvement (Brehm, 1993) or demographic characteristics that reduce either the ability or willingness to cooperate with an interview (Groves & Couper, 1998). Our findings could be interpreted to suggest that it is not interest in politics *per se*, but the *type of motivation* (ideological vs. access-seeking) that is correlated with nonresponse amongst elites.

Several findings within our data support this conclusion. That donors who contribute small amounts are more likely to respond to our lengthy questionnaire is consistent with this view about what might motivate donors who give different amounts. Individuals who are involved in politics more as a result of ideological concerns would take more interest and respond more favorably to lengthy surveys

about political attitudes and beliefs than individuals who view participation in politics more as an investment. That we find nonrespondents giving earlier (though not by much) is also consistent with this view, since the earlier contributors are largely considered to be more of the insider (access-seeking) variety. Finally, Nader donors' overall high response rate is similarly consistent, as no one can have imagined Nader would win the presidency.

Although it is not conclusive, some further evidence is available from the surveys themselves. Respondents were asked if having a candidate who is 'friendly to my industry and work' is important in their 'decision to contribute to a Presidential candidate.' Amongst the two subgroups with the lowest response rates (Bush and Dole) far fewer respondents rejected this suggestion than amongst the groups with the highest response rates (McCain and Nader). Roughly 26 percent ($n=76$) of Bush respondents and 31 percent ($n=86$) of Dole respondents indicated that these concerns were not important when deciding to contribute. However, approximately 42 percent ($n=116$) of McCain respondents and 58 percent ($n=237$) of Nader respondents rejected this statement.

In conclusion, we find in contrast to previous studies, that although respondents and nonrespondents differed along dimensions of political activity, nonrespondents tended to be somewhat more involved. However, the magnitude of what differences do exist is seldom of substantive theoretic interest, and seems to be explained by the different types of motivation individuals have for participating in politics.

We conclude with some discussion of the limitations of our study and suggestions for further analysis. First, we presented univariate comparisons to paint an overall picture of nonrespondents on various dimensions of political activity. We examined logistic regressions in attempts to identify the variables most predictive of nonresponse. However, these regressions offered little additional insights beyond the results from the univariate comparisons, since many of the predictors were highly correlated. Second, some caution should be taken in interpreting these results, as the Nader subsample did not receive a third wave of mailers (although, as noted above, this is unlikely to change many of our results as so few individuals responded to the third mailing). Finally, as this is the first analysis of differences between elite respondents and nonrespondents along dimensions of political activity, researchers should look for opportunities to confirm our findings with other surveys of elite samples.

APPENDIX: SURVEY QUESTIONS

How important are the following factors in your decision to contribute to a Presidential candidate? Candidate is friendly to my industry. Very; Somewhat; Not at all.

Annual family income: Less than \$50,000; \$50,000–\$99,000; \$100,000–\$249,000; \$250,001–\$500,000; Above \$500,000.

Have you ever done any of the following? Please check all that apply. Testified before Congress; Spoken with the President.

Education: High School or less; Some College; College Degree; Some Graduate School; Graduate Degree (respondents were asked to write in their degree if they chose the Graduate Degree option).

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